

Mark Farinas

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Technical Skills

Languages: Python, Kotlin, Dart, SQL, JavaScript, HTML/CSS, GDScript, VBA

Frameworks & Libraries: Jetpack Compose, Flutter, Ktor, JetBrains Exposed, Retrofit, Next.js, React, Tailwind CSS

Databases & Infrastructure: PostgreSQL, PostGIS, SQLite, Supabase, Microsoft Access, NocoDB, Docker

AI/ML: Ollama, Groq, RAG pipelines, Whisper AI, LLM evaluation, agent-trajectory analysis, inter-rater agreement (Cohen's κ , Krippendorff's α)

Cloud & DevOps: Cloudflare (Workers, DNS, Email Routing), Vercel, Railway.io, Firebase, GitHub Actions, DNS/TLS administration

Tools: Git, Figma

Education

MacEwan University - Edmonton, AB

Jan 2024 – Jun 2028

BSc in Computer Science, Minor in Business Studies

- **cGPA:** 3.6 / 4.0
- **Award (Apr 2026):** 1st Place - CMPT 395 Software Engineering Competition (IEEE-Sponsored) - won both the Semi-Finals and the Grand-Finals with Schedulater
- **Publication (Accepted Jul 2026):** Co-authored full research paper accepted to IEEE SmartEdu 2026 - "Unjamming Deferrals: Supporting Students Through JAMN Bot, a Mobile RAG-Grounded Policy Assistant"
- **Conference Talk (Sep 10, 2026):** Sole presenter of the paper above at IEEE SmartEdu 2026 (IEEE Smart World Congress), delivered online on behalf of four co-authors and two faculty supervisors

University of Alberta - Edmonton, AB

Sep 2016 – Apr 2021

BSc in Immunology and Infection

Experience

Cohere - Remote

Aug 2026 – Present

Data Annotation Specialist, Software Engineering (Independent Contractor)

- Evaluate model-generated code and agent trajectories against correctness, security, and edge-case rubrics; justifications become LLM training signal
- Debug AI-generated code across Python, JavaScript, Java, Go, and SQL; assess tool selection and error recovery in agent traces

Alberta Federation of Rural Water Coops - Edmonton, AB

Junior Database Developer

Sep 2026 – Present

- Own the legacy Access database's conversion to a web-based system through cutover, training, and handover
- Administrate the Water Federation website; automate form development and recurring administrative workflows

Database Administrator Intern

May 2026 – Aug 2026

- Built a re-runnable Python ETL pipeline (pyodbc to psycopg) migrating a legacy Access database to containerized PostgreSQL + NocoDB, validated by 125 fidelity tests
- Built TimeEntrySystem, a role-based Access/VBA timesheet app: three-stage approval workflow, three-layer anti-fraud audit model
- Built InsuranceRenewalSystem, an Access/VBA workflow matching five coverages by organization ID and generating 85 verification documents
- Coordinated migration rollout with stakeholders across rural Alberta water systems

Alberta Precision Labs - Edmonton, AB
COVID-19 Surveillance Research

Dec 2022 – Dec 2023

- Ran qPCR and dPCR COVID-19 surveillance across Albertan communities; daily RNA extraction and RT-PCR workflows

Popeye's Supplements - Edmonton, AB
Customer Service

Jul 2022 – May 2026

- Advised clients on supplements for athletic performance and wellness goals across four years; ran in-store demos and community education initiatives

Projects

Schedulater - Backend & Android Developer, 5-person Agile/Scrum team
Kotlin, Ktor, JetBrains Exposed, SQLite, Jetpack Compose, Docker, Ollama

Jan 2026 – Apr 2026

- Architected the Ktor + Exposed + SQLite backend: JWT role auth, room-booking engine with conflict detection and proctor load-balancing
- Built role-specific student, faculty, and admin screens in Kotlin + Jetpack Compose over Retrofit, Firebase Auth, FCM, and DataStore
- Designed automated FCM push triggers and a scheduled no-show poller in the backend service layer
- Built an Ollama RAG policy assistant: nomic-embed-text embeddings, cosine-similarity retrieval over a SQLite vector store, role-aware prompt builder
- Containerized with Docker and deployed on Railway.io with environment-driven secrets

iMapped - Chief Operating Officer; Web Core & Release Engineering
Flutter, Dart, Supabase, PostGIS, Groq, Vercel, GitHub Actions

May 2026 – Present

- Built the web core of a trip-itinerary planner live at imapped.ca  - one Flutter codebase to web and native Android, 172 tests green
- Implemented the client-side route optimizer in Dart: haversine distance, nearest-neighbour construction, 2-opt improvement
- Integrated Supabase (Postgres + PostGIS, PKCE auth, Edge Functions) behind repository interfaces, letting fakes swap for live implementations without UI changes
- Deployed a Groq (Llama 70B) recommendation re-ranker as a Supabase Edge Function, caching results to control latency and API cost
- Led UI/UX and two rounds of user research, driving a pivot from flight booking to itinerary planning
- Built as a 5-person venture in COSL 301; carried it past the course owning the GitHub organization, release process, and live site

JAMN RAG Benchmark - Evaluation design and analysis

2026

Python, LLM-as-judge, inter-rater agreement; [jamn-rag-benchmark](#) 

- Designed the evaluation behind the paper above: a 43-item benchmark with hand-authored ground truth and a three-rater human baseline (Krippendorff's $\alpha = 0.894$)
- Established a sampling-noise floor of ≈ 0.08 from an 11-point temperature ablation, and used it to retract two leaderboard claims
- Traced a published groundedness result to a judging-harness defect - the judge saw chunk IDs, not retrieved text - and published the corrected re-judge
- Released the reproducibility artifact (Zenodo DOI 10.5281/zenodo.21569635) with a `reproduce.py` regenerating every headline number offline

Retail Storefront Rescue & Migration - Solo client engagement
Cloudflare Workers, DNS/TLS, GoDaddy

Jul 2026

- Rebuilt a local retailer's storefront overnight from an unusable dev-mode capture after a multi-day outage; dependency-free static site at \$0/month
- Executed the nameserver cutover with zero downtime for the client's online store, verifying DNSSEC state before touching the legacy zone

- Configured TLS, Cloudflare Email Routing, and SPF/DKIM/DMARC (**p=reject**); added a fails-closed 18+ age gate
- Structured the handover around client ownership: client-held domain and admin access, plus a written runbook

VocabMaxxing [🔗](#) - Android, Backend & UX Lead

Feb 2026 – Jul 2026

Kotlin, Jetpack Compose, Ktor, Groq, Firebase, Figma

- Implemented the Android frontend (login, dashboard, daily practice with microphone capture) in Kotlin + Jetpack Compose
- Built the Ktor + Exposed + SQLite backend with Firebase Admin token verification; containerized on Railway.io
- Designed a Groq (Llama 3.3 70B) scoring service with a multi-dimensional rubric and prompt-injection defenses, hybridized with a morphological-stemming scorer

Additional Projects

- **Phone-Menu-Transcriber** [🔗](#) (Jun 2026) - Whisper speech-to-text with a local Qwen 3 8B extractor; Pydantic-constrained JSON as CLI and FastAPI service, `mypy --strict`, four-version CI matrix
- **Maintenance Calorie Finder** [🔗](#) (Jul 2025) - Python CLI modelling metabolic adaptation and solving for target intake with SciPy bounded optimization; 24 pytest cases
- **Personal Portfolio** [🔗](#) (Dec 2025 – Present) - Next.js, React and Tailwind with a custom design system and client-side routing
- **TiMED** (Hack4Health 2025) - co-designed a hardware-software medication management system and presented the proof-of-concept to judges